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CO1-AT1

1) A college wants to identify students who may fail in the final exam. How can Data Science Help?

A) Data Science can analyze students attendance, internal marks, assignment scores, previous exam performance, and study patterns to predict which students are at risk of failing. Machine learning models can identify high-risk students early, allowing teachers to provide extra classes, counseling, and academic support.

2) An online shopping website wants to recommend products to customers. Which use of Data Science is applied here?

A) This is an application of Recommendation systems. Data Science analyzes customer's previous purchases, search history, product views, ratings, and preferences to recommend relevant products. This improves the customer experience and can increase sales.

3) A company wants to reduce customer loss. How can Data Science support this?

A) Data Science can be used for customer churn predictions. It analyzes customer behavior, purchase history, complaints, usage patterns, and interactions to identify customers who are likely to leave. The company can then offer discounts, personalized services, or support to retain them.

4) A supermarket wants to Know which products sell more during festivals. How can Data Science Help?

A) Data Science can analyze historical sales data, festival periods, customer purchases, and product demand to identify sales patterns. It can predict which products will have high demand during festivals, helping the supermarket maintain proper stock and plan promotions.

5) A farmer wants to improve crop yield. Give one use of Data Science.

A) Data Science can be used for crop yield prediction. It can analyze soil conditions, rainfall, temperature, humidity, irrigation and previous crop data to predict the expected yield. Farme can use these predictions to improve crop selection, irrigation, and fertilizer usage.

6) A wheather app updates temperature every minute. Which facet of data is involved?

A) The facet involved is velocity. Velocity refers to the speed at which data is generated, collected, and processed. Since the weather app recieves and updates temperature data every minute, it demonstrates high data velocity.

7) Customer comments on social media are collected for analysis. What type of data is this?

A) Customer comments on social media are mainly unstructured data because they are free-form text and may contain emojis, images, hashtags, and different formats. Data Science can analyze this during Natural Language processing (NLP) to understand customer opinions and sentiment.

8) A hospital dataset contains missing age values and incorrect phone numbers. Which data quality issue is present?

A) The data has data quality issues, specifically completness and accuracy problems. Missing age values represent a completeness issue, while incorrect phone numbers represents an accuracy issue. Data cleaning and validation should be performed before analysis.

9) A company stores large files in Hadoop Distributed file System. which part of the Big Data ecosystem is used?

A) The Hadoop Distributed file System (HDFS) is the storage component of the Hadoop ecosystem. It stores very large datasets by distributing files across multiple computers or nodes, providing scalability and fault tolerance.

10) An organization wants to collect live sensor data from IOT devices. Which Big Data ecosystem component is suitable?

A) A real-time data ingestion and streaming component such as Apache Kafka is suitable. Kafka can collect and transport continues streams of data from IOT sensors effectively. This allows organizations to process real-time sensor data for monitoring and decision making.